

Notes: Taschereau-Dumouchel (RES 2026)

Cascades and Fluctuations in an Economy with an Endogenous Production

Network

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1 Big picture

1.1 Question

The paper asks: *how does a production network form endogenously when firms choose whether to operate*, and what are the implications for:

1. the **structure of the production network** (degree distributions, clustering, distances),
2. **cascades** of firm exits (propagation of shutdowns), and
3. **aggregate fluctuations** when firm-level productivities are persistent over time.

1.2 Core mechanism

A firm that operates both *creates a node* and *activates its potential input–output links*; because operating is costly (fixed overhead labor) and valuable (variety + intermediate-input complementarity), the optimal set of operating firms is **lumpy** and **clustered**, which endogenously generates cascades and business-cycle movements in network topology.

1.3 Main contributions

1. **Endogenous network formation through the extensive margin.** Potential linkages are encoded in a matrix Ω , but the *active* production network depends on which firms operate.
2. **A computational method for the combinatorial planner problem.** Choosing the operating set $\theta \in \{0, 1\}^n$ is NP-hard in general. The paper introduces a *reshaping* method: relax θ to $[0, 1]^n$ and modify the objective with shape parameters so that (i) the optimum remains at the corners and (ii) the relaxed problem can be solved reliably at large n .
3. **Cascades from endogenous shutdown decisions.** Because operating decisions are strategic complements (upstream and, under some pricing regimes, downstream), a firm exit can trigger a chain of exits. High-degree firms are less likely to exit but generate larger cascades when they do.
4. **Macro implications of network reorganization.** Network reorganization raises the *level* of output and can *dampen volatility* relative to an economy that keeps the network fixed after it is optimally designed at time 0.

Figure 1 interpretation (potential vs active network). Panel (a) plots the *potential* connections encoded in Ω . Panel (b) plots the *active* connections in the efficient allocation: a link $i \rightarrow j$ is active only if (i) $\Omega_{ij} > 0$ and (ii) both i and j operate ($\theta_i = \theta_j = 1$). This is the central modeling move: the production network is *endogenous* because the extensive-margin decision θ turns links on/off.

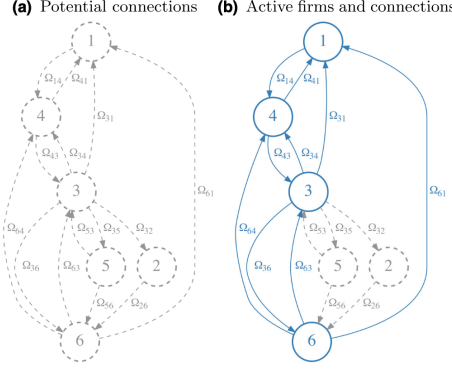


FIGURE 1
The firms' operating decisions determine the production network (a) potential connections and (b) active firms and connections

1.4 Notation map (variables that appear everywhere)

- Firms: $j \in \{1, \dots, n\}$.
- Operating decision: $\theta_j \in \{0, 1\}$ (operate or not); in the reshaped relaxation, $\theta_j \in [0, 1]$.
- Productivity: $z_j > 0$ (idiosyncratic); $A > 0$ (aggregate TFP).
- Fixed overhead labor: $f_j > 0$ (paid only if $\theta_j = 1$).
- Technology: $\alpha_j \in (0, 1)$ intermediate-input share; $\varepsilon_j > 1$ elasticity across intermediate inputs for firm j .
- Preferences: $\sigma > 1$ elasticity across consumption varieties.
- Network primitives: $\Omega_{ij} \geq 0$ indicates whether i can supply j (and how strongly).
- Allocation objects: consumption c_j , intermediate flows x_{ij} , labor l_j , output y_j .
- Dual variables in the planner: λ_j (shadow value of good j), w (shadow value of labor).
- Convenient productivity index: $q_j \equiv w/\lambda_j$ ("labor productivity" of variety j in equilibrium/planner).

2 Primitives and the efficient allocation

2.1 Household preferences

A representative household consumes the n differentiated goods $\{c_j\}_{j=1}^n$ with CES aggregator

$$C = \left(\sum_{j=1}^n \beta_j^{1/\sigma} c_j^{(\sigma-1)/\sigma} \right)^{\sigma/(\sigma-1)}, \quad \sigma > 1, \quad \beta_j > 0. \quad (1)$$

The parameter σ governs love for variety (low σ means varieties are harder to substitute).

2.2 Firm technology and potential production network

Each firm j produces output y_j using labor l_j and a CES bundle of intermediates $\{x_{ij}\}_{i=1}^n$:

$$y_j = A^{\alpha_j} (1 - \alpha_j)^{1-\alpha_j} z_j \theta_j \left(\sum_{i=1}^n \Omega_{ij}^{1/\varepsilon_j} x_{ij}^{(\varepsilon_j-1)/\varepsilon_j} \right)^{\frac{\varepsilon_j}{\varepsilon_j-1} \alpha_j} l_j^{1-\alpha_j}. \quad (2)$$

Interpretation:

- $\Omega_{ij} = 0$ means i cannot supply j (no potential link).
- The *active* network depends on θ : if i does not operate, it produces $y_i = 0$ and cannot supply anyone, so the corresponding outgoing links are inactive.
- ε_j controls how valuable input variety is for firm j .

2.3 Planner problem

The planner chooses (c, x, l) and the operating set θ to maximize C subject to goods and labor feasibility:

$$\max_{\{c_j, x_{ij}, l_j, \theta_j\}} C \quad (3)$$

$$\text{s.t.} \quad c_j + \sum_{k=1}^n x_{jk} \leq y_j \quad \forall j, \quad (4)$$

$$\sum_{j=1}^n (l_j + \theta_j f_j) \leq 1, \quad (5)$$

$$\theta_j \in \{0, 1\}, \quad c_j \geq 0, \quad x_{ij} \geq 0, \quad l_j \geq 0.$$

2.4 Solving the planner conditional on a fixed operating set θ

Fix θ and consider the remaining choice over (c, x, l) . Let λ_j be the multiplier on eq. (4) and w on eq. (5). Define the **productivity index**

$$q_j \equiv \frac{w}{\lambda_j}.$$

Key fixed point (Lemma 1 in the paper). For any fixed θ , the vector $q = (q_1, \dots, q_n)$ solves:

$$q_j = z_j \theta_j A \left(\sum_{i=1}^n \Omega_{ij} q_i^{\varepsilon_j-1} \right)^{\frac{\alpha_j}{\varepsilon_j-1}}. \quad (6)$$

This mapping makes the network channel explicit: q_j depends on upstream productivities q_i through the potential input weights Ω_{ij} and the CES curvature ε_j .

Aggregate productivity and GDP (Lemma 2 in the paper). Define the aggregate productivity index

$$Q = \left(\sum_{j=1}^n \beta_j q_j^{\sigma-1} \right)^{1/(\sigma-1)}. \quad (7)$$

Then optimal consumption/GDP can be written as

$$C = Q \left(1 - \sum_{j=1}^n \theta_j f_j \right). \quad (8)$$

This decomposition is extremely useful: for given θ , the economy looks like “effective TFP” Q times “effective labor” $(1 - \sum \theta_j f_j)$.

Economic intuition. Turning on an additional firm j has two opposing effects:

- **Benefit:** it increases Q directly (more varieties, possibly higher q_j) and indirectly (it can become an input supplier that raises other firms’ q ’s via eq. (6)).
- **Cost:** it uses overhead labor f_j , reducing the available labor term in eq. (8).

2.5 Reduced-form planner over θ (where the difficulty begins)

Plugging eqs. (6) to (8) into the planner reduces the hard part to:

$$\max_{\theta \in \{0,1\}^n} \underbrace{Q(\theta)}_{\text{network productivity via } q(\theta)} \cdot \underbrace{\left(1 - \sum_j \theta_j f_j \right)}_{\text{labor net of overhead}}.$$

This is combinatorial and generally non-convex: because $Q(\theta)$ can jump when θ changes (links appear/disappear), the objective can have many local maxima.

3 The reshaping method: turning an NP-hard search into a tractable optimization

3.1 Step 1: relax $\theta \in \{0,1\}^n$ to $\theta \in [0,1]^n$

The paper first considers a relaxed problem where firms can be “partially active” ($\theta_j \in [0,1]$). But relaxation alone is not enough: the objective can still be non-concave and can still admit interior stationary points.

$\mathcal{P}$: Original planner's problem	$\mathcal{R}$: Relaxed and reshaped problem
$\max_{\theta \in \{0,1\}^n} Q \left(1 - \sum_{j=1}^n \theta_j f_j \right) \quad (8)$	$\max_{\theta \in [0,1]^n} Q \left(1 - \sum_{j=1}^n \theta_j f_j \right) \quad (8)$
where q solves, for each $j \in \mathcal{N}$, $q_j = z_j \theta_j A \left(\sum_{i=1}^n \Omega_{ij} q_i^{\varepsilon_j-1} \right)^{\frac{\alpha_j}{\varepsilon_j-1}} \quad (6)$	where q solves, for each $j \in \mathcal{N}$, $q_j = z_j \theta_j^{a_j} A \left(\sum_{i=1}^n \Omega_{ij} \left(\theta_i^{b_{ij}} q_i \right)^{\varepsilon_j-1} \right)^{\frac{\alpha_j}{\varepsilon_j-1}} \quad (9)$

FIGURE 2
Differences between the original and reshaped problems

3.2 Step 2: reshape the mapping so that (i) corners are preserved and (ii) corner solutions are selected

Introduce shape parameters a_j and b_{ij} and replace eq. (6) by:

$$q_j = z_j \theta_j^{a_j} A \left(\sum_{i=1}^n \Omega_{ij} (\theta_i^{b_{ij}} q_i)^{\varepsilon_j-1} \right)^{\frac{\alpha_j}{\varepsilon_j-1}}. \quad (9)$$

This corresponds to a reshaped production function:

$$y_j = A^{\alpha_j} (1 - \alpha_j)^{1-\alpha_j} z_j \theta_j^{a_j} \left(\sum_{i=1}^n \Omega_{ij}^{1/\varepsilon_j} (\theta_i^{b_{ij}} x_{ij})^{(\varepsilon_j-1)/\varepsilon_j} \right)^{\frac{\varepsilon_j}{\varepsilon_j-1} \alpha_j} l_j^{1-\alpha_j}. \quad (10)$$

Corner preservation (the key accounting trick). If $\theta \in \{0,1\}^n$, then $\theta_j^{a_j} = \theta_j$ and $\theta_i^{b_{ij}} = \theta_i$ for any real a_j, b_{ij} , so the reshaped and original problems *coincide on the corners*. This is why the method can work: it changes the objective *between* the corners in order to make the correct corner easier to find.

Figure 2 interpretation (what reshaping changes, and what it does *not* change). Figure 2 is a compact schematic:

- The left side shows the original mapping $q = \Psi(q; \theta, z)$ and the induced GDP/consumption expression.
- The right side shows the reshaped mapping $\tilde{\Psi}$ in eq. (9): θ enters with exponents a_j and b_{ij} .
- Crucially, when $\theta \in \{0,1\}^n$, both problems coincide (the same corners), but the “in-between” geometry is altered to remove spurious local maxima.

Proposition 1 (corner equivalence). Any corner solution of the reshaped problem is also a solution of the original planner problem.

3.3 First-order condition for θ_j in the reshaped problem (Lemma 3)

Let λ_j be the shadow value of good j in the reshaped problem. Define the reshaped intermediate bundle X_j and its social value Λ_j as:

$$X_j = \left(\sum_{k=1}^n \theta_k^{b_{kj} \frac{\varepsilon_j - 1}{\varepsilon_j}} \Omega_{kj}^{1/\varepsilon_j} x_{kj}^{(\varepsilon_j - 1)/\varepsilon_j} \right)^{\varepsilon_j / (\varepsilon_j - 1)}, \quad (11)$$

$$\Lambda_j = \left(\sum_{k=1}^n \theta_k^{b_{kj}(\varepsilon_j - 1)} \Omega_{kj} \lambda_k^{1 - \varepsilon_j} \right)^{1/(1 - \varepsilon_j)}. \quad (12)$$

The shadow value of good j can be written as

$$\lambda_j = \frac{1}{z_j \theta_j^{a_j} A \Lambda_j^{\alpha_j} w^{1 - \alpha_j}}. \quad (13)$$

The first-order condition w.r.t. θ_j takes an *accounting* form:

$$\underbrace{a_j \frac{\lambda_j}{\theta_j} c_j}_{\text{marginal gain via consumption}} + \underbrace{\sum_k (a_j + b_{jk}) \lambda_j \frac{\theta_j^{b_{jk}(\varepsilon_j - 1)}}{\theta_j} \Omega_{jk} X_k \left(\frac{\lambda_j}{\Lambda_k} \right)^{-\varepsilon_k}}_{\text{marginal gain via intermediate links}} - \underbrace{w f_j}_{\text{marginal cost}} = \Delta \mu_j, \quad (14)$$

where $\Delta \mu_j$ collects KKT multipliers from the box constraint $\theta_j \in [0, 1]$.

How to read eq. (14). The left-hand side is “marginal benefit of raising θ_j ” minus “marginal cost”. Different choices of shape parameters change how the marginal benefit varies with θ_j .

3.4 Choosing shape parameters to force corner solutions (the $(*)$ condition)

The paper sets

$$a_j^* = \frac{1}{\sigma - 1}, \quad b_{ij}^* = \frac{1}{\varepsilon_j - 1} - \frac{1}{\sigma - 1}. \quad (15)$$

Under these values, the *direct* dependence of the marginal benefit in eq. (14) on θ_j disappears (or becomes weak in large n). As a result, KKT solutions are pushed to $\theta_j \in \{0, 1\}$.

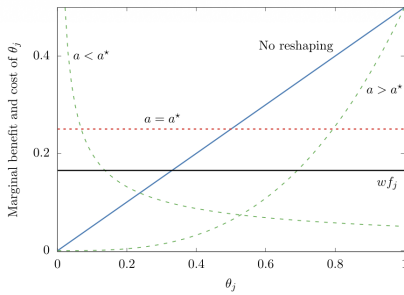


FIGURE 3
Impact of a_j on the marginal productivity of θ_j

Notes: Marginal benefit (first two terms in (12), all lines except the black one) and marginal cost (third term $w f_j$ in (12), black line) of increasing θ_j under different values of a_j . Parameters: $a_j = 0$, $\sigma_j = 3$, $\beta_j = 1/2$, $f_j = 0.15$, $C = z = w = 1$

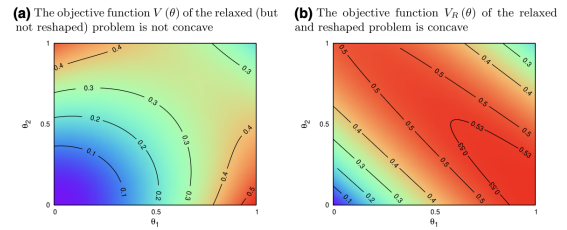


FIGURE 4

Reshaping the planner's problem in a simple economy (a) The objective function $V(\theta)$ of the relaxed (but not reshaped) problem is not concave and (b) The objective function $V_R(\theta)$ of the relaxed and reshaped problem is concave

Notes: The parameters are $n = 2$, $\alpha_j = 0.5$, $L = 1$, $f = 0.45$, $\sigma = \varepsilon_j = 3$, $c_1 = 1$, $c_2 = 0.95$, $\Omega_{ij} = 1$ for all i, j

Figure 3 intuition (marginal benefit vs marginal cost). Figure 3 in the paper plots the marginal benefit curve (first two terms in eq. (14)) under different a_j 's. At the special value $a_j^* = 1/(\sigma - 1)$, the marginal benefit is flat in θ_j , so the solution occurs at a corner once the fixed cost line is added.

TABLE 1
Testing the solution approach for small n

n	With reshaping		Without reshaping	
	Correct θ (%)	Error in C (%)	Correct θ (%)	Error in C (%)
4	100	0.000	91.5	0.502
6	>99.9	0.000	88.1	0.602
8	99.9	0.000	86.5	0.791
10	99.9	0.001	85.3	0.855
12	99.9	0.001	84.5	0.904
14	99.9	0.001	84.0	0.928

Notes: See Supplementary Appendix B for the details of the simulations. $x > 99.9\%$ indicates that $x < 100\%$ but proper rounding would yield 100.

TABLE 2
Testing the reshaping approach for a large n

n	With reshaping		Without reshaping	
	Correct θ (%)	Error in C (%)	Correct θ (%)	Error in C (%)
1,000	>99.9	<0.001	66.5	0.56

Notes: Parameters as in the calibrated economy of Section 6.2. I simulate 100 different matrices Ω and, for each Ω , draw 100 productivity vectors ε from the procedure described in Supplementary Appendix E.3 on each of them and report average results. $x < 0.001\%$ indicates that $x > 0$ but proper rounding would yield 0.

TABLE 3
Production network in the calibrated economy and in the data

	Model			Dataset	
	Calibrated	Random	Inefficient	Factset	Compustat
Power law exponents					
Indegree distribution	0.97	1.18	1.02	0.97	1.32
Outdegree distribution	0.92	1.15	0.95	0.83	2.22
Centrality distribution	1.16	1.23	1.22	0.59	0.06
Sales distribution	0.80	0.60	0.80	–	0.54
Global clustering coefficient	3.45	2.08	2.99	3.46	0.09
Average distance	2.64	3.94	2.66	4.81	1.04

Notes: To focus on the right tail, the eigenvector centrality and the sales distribution are truncated below the first quartile. Global clustering coefficients are computed on the undirected graph and multiplied by the square roots of the number of nodes. See footnote 20 for details. The average distance and the eigenvector centrality are computed on the undirected graph. "Inefficient" refers to the distorted equilibrium of Section 4. All power law exponents are computed using the estimator of Galbraith and Ibragimov (2011). Since a large fraction of firms in Factset are private, its sales data is sparse and is not reported.

3.5 Why reshaping helps numerically (Figure 4 + Tables 1–2)

Figure 4. In a two-firm example, the relaxed (but not reshaped) objective is non-concave with multiple local maxima, so gradient methods can converge to the wrong corner. After reshaping, the objective becomes concave and gradient ascent is well behaved.

Table 1 (small n). With reshaping, the algorithm recovers the correct θ essentially always and the consumption error is close to zero (fractions of a percent). Without reshaping, the algorithm frequently converges to incorrect operating sets and the welfare loss can be close to 1% even for $n = 14$.

Table 2 (large $n = 1000$). Reshaping remains accurate at scale: the reported rate of correct θ is above 99.9% with negligible consumption error, whereas without reshaping the correct- θ rate falls to about 66.5% and welfare errors become material.

4 Decentralizing the allocation: undistorted vs distorted equilibrium

4.1 Households

Given prices p_j , the household chooses $\{c_j\}$ to maximize eq. (1) subject to the budget constraint

$$\sum_{j=1}^n p_j c_j \leq w \left(1 - \sum_{j=1}^n \theta_j f_j \right) + \sum_{j=1}^n \pi_j, \quad (16)$$

where π_j are firm profits rebated to the household. The CES demand system implies

$$c_j = \beta_j \left(\frac{p_j}{P} \right)^{-\sigma} C, \quad P \equiv \left(\sum_{j=1}^n \beta_j p_j^{1-\sigma} \right)^{1/(1-\sigma)}. \quad (17)$$

4.2 Firm cost minimization (unit costs)

Given input prices $\{p_i\}$ and wage w , firm j minimizes cost subject to producing y_j with technology eq. (2). The unit cost (marginal cost) takes the fixed-point form

$$\delta_j = \frac{1}{z_j \theta_j A} \left(\sum_{i=1}^n \Omega_{ij} \delta_i^{1-\varepsilon_j} \right)^{\frac{\alpha_j}{1-\varepsilon_j}}, \quad (18)$$

and intermediate-input demand from customer k takes the CES form

$$x_{jk} = \Omega_{jk} \left(\frac{p_{jk}^x}{P_k^x} \right)^{-\varepsilon_k} X_k, \quad P_k^x \equiv \left(\sum_i \Omega_{ik} (p_{ik}^x)^{1-\varepsilon_k} \right)^{1/(1-\varepsilon_k)}. \quad (19)$$

4.3 Two market structures: where the distortion comes from

The paper contrasts:

1. an **undistorted equilibrium** that can be decentralized (monopolistic competition with appropriate subsidies/taxes),
2. a **distorted equilibrium** where *customers have bargaining power over intermediate-input prices* and set $p_{jk}^x = \delta_j$ (supplier marginal cost), preventing suppliers from extracting surplus from being used as intermediates.

Model objects: profits under the two regimes (paper Eqs. (19), (23), and (25)). Let p_j^c be the price to consumers and p_{jk}^x the price at which j sells to firm k .

- **General profit accounting with intermediate-sale wedges (Eq. (19)).** With a wedge/subsidy s_{jk}^x on intermediate sales,

$$\pi_j = p_j^c c_j + \sum_{k=1}^n (1 + s_{jk}^x) p_{jk}^x x_{jk} - \sum_{i=1}^n p_{ij}^x x_{ij} - w l_j - w f_j \theta_j. \quad (20)$$

- **Undistorted equilibrium profit (Eq. (23)).** In the undistorted equilibrium, markups apply both to consumer sales and to intermediate sales (reflecting CES demand elasticities):

$$\pi_j^u = \frac{\sigma}{\sigma-1} \delta_j c_j + \sum_{k=1}^n \frac{\varepsilon_k}{\varepsilon_k-1} \delta_j x_{jk} - \sum_{i=1}^n \delta_i x_{ij} - w l_j - w f_j \theta_j. \quad (21)$$

So operating as a supplier is privately valuable: intermediate sales generate surplus through the markup factor $\varepsilon_k/(\varepsilon_k-1)$.

- **Distorted equilibrium profit (Eq. (25)).** In the distorted equilibrium, customers set intermediate prices equal to the supplier's marginal cost ($p_{jk}^x = \delta_j$), eliminating intermediate-

sale markups:

$$\pi_j^d = \frac{\sigma}{\sigma - 1} \delta_j c_j + \sum_{k=1}^n \delta_j x_{jk} - \sum_{i=1}^n \delta_i x_{ij} - w l_j - w f_j \theta_j. \quad (22)$$

The key difference is that being an intermediate supplier no longer generates a markup-based surplus, weakening incentives to keep “upstream” varieties active.

This is the sense in which the distortion is *an entry/operation distortion*: conditional on θ , allocation is efficient (Lemma 4), but the surplus mapping from intermediates to profits is altered, so the equilibrium θ differs from the efficient θ .

A key lemma (Lemma 4) shows that *conditional on a given θ* , both equilibria implement the efficient within- θ allocation, and equilibrium prices match the planner’s shadow values (up to numeraire). Therefore, inefficiency can only come from **entry/operation decisions** (the choice of θ).

4.4 Entry decisions and their characterization via reshaping

The paper shows that equilibrium entry decisions can also be characterized by a reshaped problem.

Undistorted equilibrium (Proposition 4). If θ satisfies the first-order conditions of the reshaped planner with (a^*, b^*) in eq. (15), then θ is an undistorted equilibrium operating set.

Distorted equilibrium (Proposition 5). Under customer pricing power in intermediate markets (intermediate prices at marginal cost), the relevant reshaping parameters become

$$a_j^d = \frac{1}{\sigma - 1}, \quad b_{ij}^d = -\frac{1}{\sigma - 1}. \quad (23)$$

If θ satisfies the reshaped first-order conditions under eq. (23), it is a distorted equilibrium operating set.

Economic meaning of the distortion. In the distorted equilibrium, suppliers do not internalize the benefit of operating *as intermediates* (they cannot charge markups on intermediate sales). This pushes the economy toward:

- fewer highly connected supplier hubs,
- less clustering,
- lower productivity and GDP (quantitatively: about 0.9% lower in the calibration).

5 Mechanisms: complementarities, clustering, cascades, and macro

5.1 Upstream and downstream complementarities (Proposition 6)

Consider a partial-equilibrium view where a firm contemplates operating, taking other firms' operation fixed.

Upstream complementarity. If an upstream supplier operates, the customer can access a larger intermediate bundle at lower unit cost, which raises the customer's profitability and encourages entry.

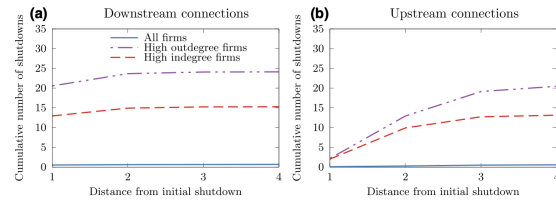
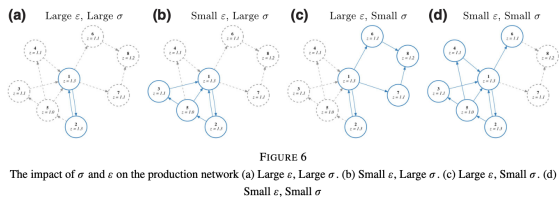
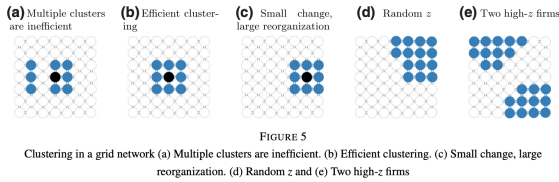
Downstream complementarity. If a downstream customer operates, it increases intermediate demand for suppliers. *This channel matters in the undistorted equilibrium (suppliers earn surplus on intermediate sales), but is muted in the distorted equilibrium when intermediate prices are set at marginal cost.*

5.2 Why clustering is efficient (Propositions 7–8 + Figure 5)

Proposition 7 (monotonicity). More productive firms and firms with better positions in the potential network are more likely to operate in the efficient allocation.

Proposition 8 (connections within a group). The value of operating a group of firms increases with the number of potential connections among them. Hence, the planner prefers **clusters** of operating firms with many internal links, because clusters create thick intermediate-input possibilities and mutually support operation decisions.

Figure 5 interpretation. In a grid-like potential network, the efficient allocation concentrates activity into one connected cluster rather than multiple separated clusters. A small productivity change can trigger a large reorganization of which region of the grid becomes active—a visual for the “lumpy” extensive-margin nature of network formation.



Notes: Cumulative number of exits at different distances from the shuttered firm. “High degree” refers to the firms above the 99th percentile. Simulations of 100 randomly drawn matrices Ω , for each of which 1,000 cascades are created

5.3 How σ and ε shape the network (Figure 6)

Figure 6 varies final-demand substitutability σ and intermediate-input substitutability ε :

- When σ and ε are **large** (inputs/varieties are easy to substitute), concentration is stronger: only the most productive firms need to operate, and the active network is smaller.
- When σ and/or ε are **small** (strong complementarities / strong love for variety), the planner has stronger incentives to activate a richer, more connected network and to build dense clusters.

5.4 Cascades of exits (Lemmas 5–6, Figure 9, Figure 10, Table 4)

Core complementarity inequality (Lemma 5). Let U_{kj} be the contribution of neighbor k to firm j 's intermediate bundle. Lemma 5 shows an inequality of the form:

$$\sum_{k: j \rightarrow k} \beta_k U_{kj} \mathbf{1}\{k \text{ operates}\} > \left(\frac{\varepsilon}{\sigma}\right) \sum_{k: j \rightarrow k} \beta_k U_{kj}, \quad (24)$$

which captures strategic complementarity: operating neighbors raise the incremental gain from operating j . If j exits, some neighbors can become unprofitable and exit too, generating a cascade.

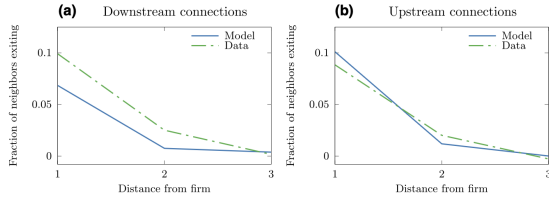


FIGURE 10
Cascades of firm shutdowns in the model and in the data (a) Downstream connections and (b) Upstream connections
Notes: Factset data. Estimated coefficients from regressing the fraction of exiting neighbours on whether a firm exits. Indegree and outdegree controls are included. The distance is the smallest number of connections between two firms

TABLE 4 High-degree firms are more resilient but create larger cascades				
	Size of cascades		Probability of exit	
	Data	Model	Data (%)	Model (%)
Average firm	0.9	1.1	11.8	11.3
High-degree firm	3.0	4.3	2.5	1.7

Notes: "High degree firms" are above the 90th percentile of the degree distribution. "Size of cascades" is the sum of exiting firms up to the second neighbours downstream and upstream, computed by multiplying the regression coefficients in Figure 10 by the number of neighbours at the corresponding distance.

Degree and cascade size (Lemma 6). The larger the set of potential connections of a firm, the stronger the complementarity effect, so high-degree firms generate larger cascades *conditional on exiting*.

Figure 9 interpretation (simulated cascades). The paper runs counterfactual cascades by shutting down one operating firm (setting its z to zero) and recomputing the efficient allocation. Cascades are mostly **downstream**: losing an upstream supplier reduces customers' productivity and can trigger further exits. High-degree originators generate much larger cumulative cascades than average firms.

Figure 10 interpretation (cascades in data vs model). The empirical object is a distance- d neighbor exit rate around an exiting firm. For downstream neighbors, the paper estimates (Eq.

(31)):

$$\text{FracExit}_{j,t+1}^{\text{down}}(d) = \alpha_d + \beta_d^{\text{down}} \mathbf{1}\{j \text{ exits at } t\} + \gamma_d^{\text{down}} \text{InDeg}_{j,t} + \delta_d^{\text{down}} \text{OutDeg}_{j,t} + \varepsilon_{j,t+1}^{\text{down}}(d), \quad (25)$$

and analogously for upstream neighbors (Eq. (32)):

$$\text{FracExit}_{j,t+1}^{\text{up}}(d) = \alpha_d + \beta_d^{\text{up}} \mathbf{1}\{j \text{ exits at } t\} + \gamma_d^{\text{up}} \text{InDeg}_{j,t} + \delta_d^{\text{up}} \text{OutDeg}_{j,t} + \varepsilon_{j,t+1}^{\text{up}}(d). \quad (26)$$

The coefficients β_d trace how exit spillovers decay with network distance.

Using Factset data, the paper estimates how exit of a firm predicts the fraction of exiting neighbors at various network distances, separately downstream and upstream. The coefficients decay with distance, and the model reproduces this qualitative pattern.

Table 4 interpretation (degree, cascades, and resilience). High-degree firms are **less likely to exit**, but **create larger cascades** when they do. The model matches both margins: it generates rare exits of hub firms and sizable systemic spillovers conditional on hub exit.

5.5 Aggregate implications of endogenous reorganization (Propositions 9–11, Table 5, Figure 11)

Continuity of GDP (Proposition 9). Even though θ is discrete, the value function $C(z)$ is continuous in the productivity vector z . Intuition: the planner can always keep the old network if switching would lower GDP, preventing downward jumps in the optimum.

Hulten’s theorem “almost everywhere” (Proposition 10). At productivity vectors where the optimal θ does not change locally,

$$\frac{\partial \log C}{\partial \log z_j} = \frac{\lambda_j y_j}{C},$$

i.e. the Domar-weight logic applies. Discrete network changes matter when θ switches.

TABLE 5
Correlation between network moments and GDP

	Model		Dataset		
	Calibrated	Inefficient	Factset	Compustat	
				CF	AHRS
Power law exponents					
Indegree distribution	−0.53	−0.18	−0.87	−0.12	−0.35
Outdegree distribution	−0.63	−0.46	−0.97	−0.11	−0.31
Centrality distribution	−0.10	0.03	−0.15	−0.39	0.30
Sales distribution	−0.44	−0.38	−	−0.24	0.04
Global clustering coefficient	0.60	0.40	0.76	0.11	0.19
Average distance	−0.82	−0.65	−0.69	0.00	0.18

Notes: All time series are in logs. In the data, output is annual real GDP, detrended linearly in sample. Since there are only 13 years in the Factset data we use the [Congressional Budget Office \(2003\)](#) 10-year projection for real GDP growth at the beginning of the sample in 2003 (2.58%) to detrend the series. To focus on the right tail, the eigenvector centrality and sales distributions are truncated below the first quartile. The global clustering coefficient, the average distance and the eigenvector centrality are computed on the undirected graph. Since Factset covers many private firms, its sales data is sparse and is not reported. “Inefficient” refers to the distorted equilibrium of Section 4.

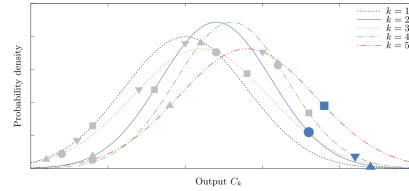


FIGURE 11
Distribution of output C_k under different networks k

Network flexibility amplifies good shocks and mitigates bad shocks (Proposition 11).

Let $\theta^*(z)$ be the efficient operating set under z . Then for a change $z \rightarrow z'$,

$$C(\theta^*(z'), z') - C(\theta^*(z), z) \geq C(\theta^*(z), z') - C(\theta^*(z), z).$$

Interpretation: allowing θ to adjust can only improve the economy’s response.

Business-cycle comovement of network moments (Table 5). In the calibration, GDP is positively correlated with clustering and negatively correlated with power-law exponents and distances, implying:

- booms feature **thicker right tails** (more hub-like networks),
- **more clustering** and **shorter distances**.

These patterns are strongest in Factset data, consistent with the model’s logic of “building clusters around well-positioned, high- z firms”.

Output level and volatility (Figure 11). Compare: (i) a flexible economy that can reorganize the network every period, versus (ii) an economy that chooses an efficient network at $t = 0$ and then keeps it fixed. The paper finds that output is **about 11% lower** and **about 17% more volatile** under the fixed network. Figure 11 interprets this as an “option value” of reorganization: the planner effectively chooses the best network among many candidates for each z .

6 Quantitative implementation and empirical evidence

6.1 Data (Section 6.1 + Figure 7)

- **Factset Revere (2003–2016):** annual firm-level supply-chain relationships collected from filings and other sources. In an average year: $\sim 13,000$ firms and $> 40,000$ links. Roughly 40% of link destructions coincide with supplier/customer shutdown.
- **Compustat (customers $> 10\%$ of sales):** the 10% disclosure threshold truncates the right tail of outdegree (few firms have > 10 observed customers).
- **CF and AHRS cleaned Compustat networks:** shorter coverage per year (about 1,300 firms and 1,500 links), but longer sample periods; useful to study business-cycle evolution.

Figure 7 visualizes the Factset network (node size relates to number of connections; darker indicates higher local clustering).

6.2 Calibration (Section 6.2 and footnote 27)

The model is static, but the quantitative exercise introduces persistence in firm productivity to generate a time series.

- **Idiosyncratic productivity:** each firm follows an independent AR(1) in logs with persistence ρ_z and stationary s.d. σ_z . The paper sets $\rho_z = 0.935$ and $\sigma_z = 0.44$ to match empirical persistence/dispersion of measured TFP.
- **Aggregate productivity:** $A = 1$.
- **Overhead labor:** choose f so that total overhead is 5% of labor: $nf = 0.05$.
- **Technology and preferences:** $\alpha = 0.5$ and $\sigma = \varepsilon = 5$ in the baseline.
- **Potential network Ω :** the in/out degrees (x^{in}, x^{out}) are drawn from a bivariate power law of the first kind:

$$\Pr(x^{in}, x^{out}) = \xi(\xi - 1) (x^{in} + x^{out} - 1)^{-\xi-1}.$$

The parameter ξ is chosen so that the *active indegree distribution* in the model matches the Factset power-law exponent (0.97).

- **Simulation:** draw 20 independent Ω 's; for each, simulate 100 periods; report averages.

6.3 Fit to network moments and firm size moments (Table 3 + Figure 8)

Table 3: what it shows. The table compares six moments between:

- the calibrated model (efficient allocation),
- a random benchmark,
- the distorted equilibrium (“inefficient”),
- Factset and Compustat-based datasets.

The moments are: power-law exponents of indegree, outdegree, eigenvector centrality, and sales; plus global clustering and average distance.

Interpretation of the main comparisons.

- The calibrated model matches Factset **indegree** and **clustering** particularly well.
- Relative to a random Ω , the efficient allocation generates **thicker tails** and **more clustering**: efficiency endogenously creates hubs and clusters.
- The distorted equilibrium features **fewer highly connected firms** and **less clustering** than the efficient allocation (consistent with intermediate-market surplus not being internalized).

- Compustat outdegree tails are much thinner due to the 10% reporting threshold (a measurement issue rather than a theory failure).



FIGURE 7
2016 Factset Reverse U.S. firm-level production network

Notes: Vector image; zoom in. Each circle is a firm, and the size of the circle reflects the number of connections. Darker colours denote higher local clustering coefficients. In 2016, Walmart had the largest indegree (448) and Microsoft had the largest outdegree (332). Image generated using Gephi with the Yifan Hu layout. There are 20,702 firms and 62,474 links

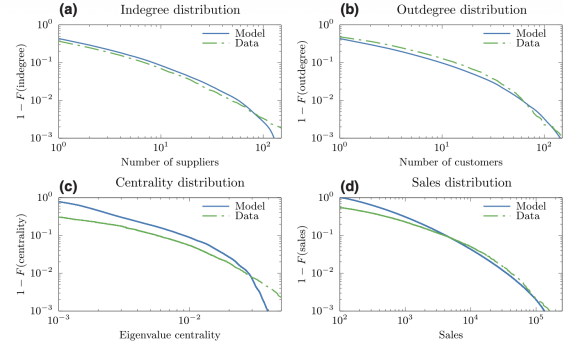


FIGURE 8

Distributions in the model and the data (a) Indegree distribution. (b) Outdegree distribution. (c) Centrality distribution. (d) Sales distribution

Notes: Efficient allocation compared to the data. The data indegree, outdegree, and eigenvector centrality distributions are from Factset in 2016 (last year in Factset). The data sales distribution is from Compustat (2016) in 2004 (last year in CF). Eigenvector centralities from the undirected graphs. Model sales and centralities are normalized to match their data averages

Figure 8: distributional fit. The figure overlays model and data distributions for indegree, outdegree, centrality, and sales. The calibrated model tracks the broad shape and tail behavior of degree distributions in Factset; mismatches are larger for centrality (which depends on indirect connections and thus is sensitive to network construction).

TABLE 3
Production network in the calibrated economy and in the data

	Model			Dataset	
	Calibrated	Random	Inefficient	Factset	Compustat
				CF	AHRS
Power law exponents					
Indegree distribution	0.97	1.18	1.02	0.97	1.32
Outdegree distribution	0.92	1.15	0.95	0.83	2.22
Centrality distribution	1.16	1.23	1.22	0.59	0.06
Sales distribution	0.80	0.60	0.80	—	0.54
Global clustering coefficient	3.45	2.08	2.99	3.46	0.09
Average distance	2.64	3.04	2.66	4.81	1.04

Notes: To focus on the right tail, the eigenvector centrality and the sales distribution are truncated below the first quartile. Global clustering coefficients are computed on the undirected graph and multiplied by the square roots of the number of nodes. See footnote 30 for details. The average distance and the eigenvector centrality are computed on the undirected graph. “Inefficient” refers to the distorted equilibrium of Section 4. All power law exponents are computed using the estimator of Gabaix and Ibragimov (2011). Since a large fraction of firms in Factset are private, its sales data is sparse and is not reported.

TABLE 4
High-degree firms are more resilient but create larger cascades

	Size of cascades		Probability of exit	
	Data	Model	Data (%)	Model (%)
Average firm	0.9	1.1	11.8	11.3
High-degree firm	3.0	4.3	2.5	1.7

Notes: “High degree firms” are above the 90th percentile of the degree distribution. “Size of cascades” is the sum of exiting firms up to the second neighbours downstream and upstream, computed by multiplying the regression coefficients in Figure 10 by the number of neighbours at the corresponding distance.

6.4 Cascades and exit spillovers (Figures 9–10 + Table 4)

- Figure 9 documents that cascades are larger downstream than upstream and are especially large for high-degree originators.
- Figure 10 estimates spillovers in data (Factset) and compares them to the model; both show decay with distance.
- Table 4 summarizes two facts in both data and model: hubs exit rarely but generate larger cascades conditional on exiting.

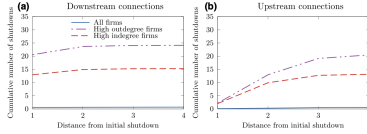


FIGURE 9
Cumulative cascades by degree of originator (a) Downstream connections and (b) upstream connections
Notes: Cumulative number of exits at different distances from the distressed firm. “High degree” refers to the firms above the 99th percentile. Simulations of 100 randomly drawn matrices Ω , for each of which 1,000 cascades are created

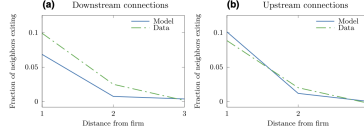


FIGURE 10
Cascades of firm shutdowns in the model and in the data (a) Downstream connections and (b) Upstream connections
Notes: Fictive data, estimated coefficients from regressing the fraction of exiting neighbors on whether a firm exits. In-degree and out-degree controls are included. The distance is the smallest number of connections between two firms

TABLE 5 Correlation between network moments and GDP					
	Model			Dataset	
	Calibrated	Insufficient	Factnet	Computat	AHRS
Power law exponents				CF	AHRS
Indegree distribution	-0.53	-0.18	-0.87	-0.12	-0.35
Outdegree distribution	-0.63	-0.46	-0.97	-0.11	-0.31
Centrality distribution	-0.10	0.03	-0.15	-0.39	0.30
Sales distribution	-0.44	-0.38	-	-0.24	0.04
Global clustering coefficient	0.60	0.40	0.76	0.11	0.19
Average distance	-0.82	-0.65	-0.69	0.00	0.18

Notes: All time series are in logs. In the data, output is annual real GDP, detrended linearly in sample. Since there are only 13 years in the Factnet data we use the Congressional Budget Office (2009) 10-year projection for real GDP growth at the beginning of the sample in 2003 (2.58%) to detrend the series. To focus on the right tail, the eigenvector centrality and sales distributions are truncated below the first quartile. The global clustering coefficient, the average distance and the eigenvector centrality are computed on the undirected graph. Since Factnet covers many private firms, its sales data is sparse and is not reported. “Insufficient” refers to the distorted equilibrium of Section 4.

6.5 Business-cycle movements in network structure (Table 5)

Table 5 reports correlations between GDP and network moments. The main takeaway is that expansions coincide with:

- thicker right tails in degree/size distributions,
- more clustering,
- shorter average path lengths.

This is exactly what the model predicts when the planner chooses to build dense clusters around temporarily productive firms.

6.6 How much does market power matter quantitatively? (Section 6.7)

In the distorted equilibrium:

- degree/centrality/size exponents are slightly larger (thinner tails),
- clustering is lower and distances are larger,
- GDP is about 0.9% lower and volatility is about 0.2% higher than in the efficient allocation.

So this particular pricing distortion matters, but the core endogenous-network mechanism remains dominant in the calibration.

7 Conclusion

7.1 Summary

This paper develops a theory of **endogenous production networks** driven by the **extensive margin of production**. Firms are connected through input–output linkages, but operating requires paying a fixed cost. As aggregate or idiosyncratic conditions deteriorate, some firms optimally shut down; their exit severs input–output links and endogenously reshapes the network.

The argument proceeds in a tight sequence:

1. **A micro-founded mechanism for network formation via entry/exit.** Firms use a CES input bundle, so additional input varieties reduce marginal costs. Consequently, the decision to operate is *complementary* across connected firms: if my suppliers operate, I become more

productive and am more likely to operate; if my customers operate, my output is more valuable and I am more likely to operate.

2. **Complementarities imply clustering and cascades in the efficient allocation.** Because operating decisions are strategic complements across neighbours, the efficient network endogenously features *clusters* of operating firms. When a firm shuts down, its disappearance weakens both its customers (who lose an input) and its suppliers (whose product becomes less useful), so exit can trigger **cascades** propagating upstream and downstream.
3. **A computational contribution: reshaping solves a hard planner problem.** The planner’s problem is non-convex (binary operation decisions) and non-concave (fixed costs create increasing returns). The paper proposes a **reshaping** transformation that relaxes the binary constraint and modifies the objective off the corners in a way that (under sufficient conditions) preserves the original optimum, making the problem tractable at large scale.
4. **Quantitative discipline using U.S. production-network data.** A calibration based on U.S. firm-level link data matches key network statistics. The efficient network exhibits thicker right tails (more highly connected firms) and stronger clustering than a neutral random benchmark (Table 3), consistent with the planner exploiting operating complementarities.
5. **Cascades: hubs are robust, but dangerous when they fail.** In the data, firms above the 90th percentile of degree generate cascades about *three times* larger than the average firm (Table 4: cascade size ≈ 3.0 vs 0.9), yet are much less likely to exit (Table 4: exit probability $\approx 2.5\%$ vs 11.8%). The calibrated model reproduces the same qualitative pattern.
6. **Business-cycle implications: recessions feature more dispersed networks.** In the model and the data, expansions are periods with (i) thicker tails of the degree distributions (more hubs), (ii) higher clustering, and (iii) shorter average distances (Table 5). Recessions are periods in which creating/maintaining dense clusters is too costly (e.g. because key firms are hit by adverse shocks), so the network becomes more diffuse.
7. **Endogenous reorganization matters for macro aggregates.** Comparing the benchmark economy (network reorganizes with shocks) to a counterfactual where the network is fixed, the paper finds that a fixed network lowers average GDP by about **11%** and makes GDP about **17% more volatile**. Intuitively, allowing the network to adapt lets the planner “route around” adverse idiosyncratic shocks by selecting a more productive configuration.
8. **Beyond efficiency: market power changes the network but not the core mechanism.** The paper also studies a distorted equilibrium with market power. The distorted network features fewer highly connected firms and weaker clustering relative to the efficient benchmark (Table 3), and is somewhat more rigid over the cycle (Table 5). Quantitatively, the gap in aggregates is modest: GDP is about **0.9%** lower and **0.2%** more volatile than under efficiency.

7.2 Economic intuition: why extensive-margin network formation creates clusters and cascades

The paper’s core intuition can be summarized in three steps.

Step 1: input variety acts like local productivity. Under CES aggregation of intermediates, gaining access to an additional active supplier reduces marginal costs. Thus, conditional on prices, a firm becomes effectively “more productive” when more of its suppliers operate.

Step 2: operating decisions are locally complementary. Because operating is more attractive when neighbours operate, firms tend to *operate together*. This generates endogenous **clustering**: dense groups of mutually beneficial supplier–customer relationships.

Step 3: cascades are the flip side of complementarity. If one firm exits, it removes an input from its customers and reduces demand/value for its suppliers. That weakens neighbours, which may exit as well; the mechanism iterates, producing **cascades** that propagate along the network. Highly connected firms are less likely to exit (they have many alternative suppliers/customers), but if they do exit, they affect many partners and can trigger larger cascades (Table 4).

7.3 Macro intuition: why endogenous reorganization can *dampen* aggregate fluctuations

A useful lens is the paper’s “envelope” representation of output: the planner effectively chooses, for each shock vector, the best operating set (hence network) among many feasible networks. This delivers two macro implications:

- **Endogenous reorganization provides a stabilization margin.** When a region of the network is hit by adverse idiosyncratic shocks, the planner reallocates activity toward other parts of the network where complementarities can be leveraged more cheaply. This selection across configurations reduces the translation of micro shocks into aggregate volatility (GDP is $\approx 17\%$ more volatile if the network is held fixed).
- **The cycle shifts the network’s shape.** In booms, it is worthwhile to pay fixed costs broadly and build dense clusters around well-positioned firms, producing thick-tailed degree distributions and high clustering. In recessions, those fixed costs are harder to justify for marginal firms, so the network becomes more dispersed with fewer hubs (Table 5).

7.4 Bottom line and open directions

- **Bottom line:** endogenous entry/exit is a powerful and empirically relevant mechanism for production-network formation, clustering, cascades, and cyclical variation in network structure.

- **Methodological takeaway:** reshaping provides a practical way to solve otherwise intractable non-convex network-formation problems.
- **Open direction (as emphasized in the conclusion):** market power is only one distortion; other externalities, coordination failures, and frictions could matter quantitatively. A natural next step is to evaluate which distortions are quantitatively important and what policies (e.g., subsidies, bailouts, coordination devices) could improve welfare in richer distorted environments.

7.5 Takeaway

When firms must pay fixed costs to operate in an input–output economy, operating decisions become complementary across neighbours, so the efficient production network endogenously forms clusters and can generate cascades of shutdowns; the resulting network structure is procyclical (dense and hub-heavy in booms, dispersed in recessions), and allowing the network to reorganize in response to idiosyncratic shocks substantially raises average GDP and lowers volatility relative to a fixed-network counterfactual.